

# Revised Research Plan: ActiStruct-nebwalk Transition-Metal Complex Reliability Benchmark

## Working Title

**ActiStruct-TMC: A Reliability-Aware Active-Learning Mini-Benchmark for DFT-Guided Transition-Metal Complex Optimization**

## Core Scientific Position

The objective is not to claim that ActiStruct or nebwalk replaces electronic-structure calculations. The objective is to demonstrate that ActiStruct can organize, evaluate, and reduce unnecessary DFT calculations by combining:

- structured candidate generation,
- Quantum ESPRESSO input/output handling,
- convergence and failure tracking,
- uncertainty-aware surrogate modeling,
- active-learning-based candidate selection,
- conservative validation against literature or benchmark reference data.

nebwalk will be used only as a secondary proof-of-concept for one molecular pathway, not as the main engine.

## Safe Scientific Claim

ActiStruct provides a reproducible framework for reliability-aware active learning in DFT-guided transition-metal complex studies. In a small benchmark of neutral organometallic complexes, it can prioritize informative geometry evaluations, quantify uncertainty, record failure modes, and support conservative DFT-validated conclusions.

## Unsafe Claims to Avoid

Do not claim:

- ActiStruct solves transition-metal chemistry.
- ActiStruct replaces DFT.
- nebwalk can accurately model all transition-metal reaction pathways.
- MLIP/MACE gives final-quality organometallic barriers.
- Dr. Kulik will be impressed.
- The workflow has minimal error without quantified validation.
- The workflow discovers new catalysts at this stage.

## Revised Material Selection

### Primary Benchmark Set: Start Here

| Priority | Complex                                            | Reason                                                                                      | Tool Role                                                        |
|----------|----------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1        | Ferrocene,<br>$\text{Fe}(\text{C}_5\text{H}_5)_2$  | Neutral, classic organometallic, closed-shell, useful for geometry and conformer comparison | ActiStruct main benchmark; nebwalk optional for Cp-ring rotation |
| 2        | Nickel tetracarbonyl,<br>$\text{Ni}(\text{CO})_4$  | Neutral, simple tetrahedral 18-electron complex, good sanity check                          | ActiStruct geometry benchmark                                    |
| 3        | Chromium hexacarbonyl,<br>$\text{Cr}(\text{CO})_6$ | Neutral, octahedral carbonyl complex, good ligand-field geometry benchmark                  | ActiStruct geometry benchmark                                    |
| 4        | Iron pentacarbonyl,<br>$\text{Fe}(\text{CO})_5$    | Neutral, trigonal-bipyramidal carbonyl, useful for fluxional behavior                       | ActiStruct geometry benchmark; nebwalk pathway candidate         |

These four systems are the best starting set because they are neutral, relatively well-studied, chemically recognizable, and less dangerous than highly charged or strongly open-shell systems.

### Secondary / Stretch Set

Use only after the primary benchmark works.

| Complex                                                 | Use Case                                         | Risk        |
|---------------------------------------------------------|--------------------------------------------------|-------------|
| $\text{V}(\text{CO})_6$                                 | Spin-polarized open-shell test                   | Medium-high |
| $[(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3]$ | Piano-stool organometallic geometry/pathway test | Medium      |
| $\text{TiCl}_4$                                         | Neutral inorganic control system                 | Low-medium  |
| 1–2 SSE17 systems                                       | Spin-state benchmark extension                   | High        |

SSE17 should be treated as a later validation extension, not the first benchmark. Spin-state energetics are strongly method-dependent and should not be used before the basic workflow is reproducible.

### Systems to Deprioritize for the First Demo

Do not start with:

- $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- $[\text{Co}(\text{NH}_3)_6]^{3+}$
- $[\text{Fe}(\text{CN})_6]^{4-}$

Reason: these are charged complexes. With plane-wave periodic DFT, charged molecular systems require careful treatment of background charge, vacuum size, electrostatic corrections, solvation/

counterion context, and spin/charge-state validation. They are scientifically important but not ideal for the first professor-facing demo.

## ActiStruct Benchmark Design

### Candidate Variables

For each neutral complex, generate small, chemically reasonable structural perturbations:

#### Ferrocene

- Fe-Cp centroid distance
- Cp ring rotation angle
- eclipsed vs staggered conformer
- small Fe-C distance perturbations

#### Ni(CO)<sub>4</sub>

- Ni-C distance
- C-O distance
- tetrahedral angle perturbation

#### Cr(CO)<sub>6</sub>

- Cr-C distance
- C-O distance
- octahedral distortion mode
- axial/equatorial perturbation

#### Fe(CO)<sub>5</sub>

- axial Fe-C distance
- equatorial Fe-C distance
- axial/equatorial angle
- Berry-pseudorotation-like distortion coordinate

## DFT Strategy

Use one consistent DFT setup for the first benchmark:

- Quantum ESPRESSO
- PBE or PBE-D3 if dispersion is needed
- PAW/USPP pseudopotentials from one family where possible
- large cubic cell for isolated molecules
- Gamma-point calculation
- sufficient vacuum spacing
- spin setting explicitly recorded
- total charge explicitly recorded
- convergence thresholds recorded
- smearing avoided for closed-shell molecules unless necessary
- failed/non-converged calculations kept in the dataset

For professor-facing output, every calculation row must record:

- complex name
- charge
- multiplicity/spin setting
- functional
- pseudopotentials
- ecutwfc
- ecutrho
- cell size
- convergence status
- final energy
- key bond distances
- max force
- wall time
- warning/failure reason
- reliability label

## ActiStruct ML/AL Task

The first task is not large-scale prediction. The first task is local active learning around known transition-metal complexes.

### Model Targets

Start with:

- relative energy within each complex family,
- convergence/failure risk,
- geometry reliability label.

Do not train one global model across unrelated complexes unless the dataset becomes large enough.

### Uncertainty

Use:

- ensemble standard deviation,
- bootstrap uncertainty,
- split conformal intervals only if the dataset size is meaningful.

Report uncertainty honestly. Do not claim calibrated uncertainty unless coverage is measured.

### Acquisition Function

For each complex family:

```
score = predicted_relative_energy - alpha * uncertainty + beta * failure_risk -  
gamma * diversity_bonus
```

Interpretation:

- low predicted relative energy is useful,
- high uncertainty can be informative,
- high failure risk is penalized,
- diversity avoids repeatedly selecting near-identical perturbations.

## nebwalk Role

nebwalk should be used only after ActiStruct produces reliable endpoints.

Recommended first nebwalk pathway:

### Option A: Ferrocene Cp-ring rotation

Endpoint 1: eclipsed conformer

Endpoint 2: staggered conformer

Goal: demonstrate molecular pathway setup and relative rotation barrier qualitatively.

### Option B: $\text{Fe}(\text{CO})_5$ Berry pseudorotation

Endpoint 1: trigonal-bipyramidal arrangement

Endpoint 2: permuted axial/equatorial arrangement

Goal: demonstrate a fluxional organometallic pathway.

Do not start with CO dissociation as the first nebwalk demo because bond-breaking pathways are more sensitive to electronic state, box size, spin state, and fragment reference energies.

## Validation Metrics

For each system, report:

### Geometry

- metal–ligand bond length error
- ligand internal bond length error
- key angle error
- RMSD after alignment if possible

### Energetics

- relative energy trends within each family
- conformer energy difference if available
- barrier only for the one nebwalk pathway

### Reliability

- number of calculations attempted
- number converged
- number failed
- number selected by active learning

- number avoided due to predicted failure risk
- wall-time comparison if available

## ML/AL

- MAE/RMSE only if dataset size is meaningful
- uncertainty vs actual error
- active-learning selection vs random selection
- selected candidate rationale

## Report Structure

Create a short technical report:

### Title:

ActiStruct-TMC: Reliability-Aware Active Learning for DFT-Guided Transition-Metal Complex Optimization

### Sections:

1. Motivation
2. Why transition-metal complexes are challenging
3. Why reliability-aware DFT workflows matter
4. Selected benchmark systems
5. ActiStruct workflow
6. DFT settings and reproducibility
7. Candidate-generation strategy
8. ML and uncertainty method
9. Active-learning selection logic
10. Benchmark results
11. nebwalk pathway demonstration
12. Failure and reliability analysis
13. Limitations
14. Next steps

## Final Outreach Positioning

Use this wording:

I am developing ActiStruct as a small reliability-aware active-learning workflow for DFT-guided transition-metal complex studies. The current prototype does not replace DFT; it organizes DFT data, tracks convergence and failure modes, estimates uncertainty, and selects informative next calculations for validation. I tested the idea on a small set of neutral organometallic benchmark complexes and used nebwalk only as a single-pathway demonstration.

## Minimum Evidence Before Contacting Dr. Kulik

Do not contact her until you have:

- clean ActiStruct GitHub repository,

- 4 neutral benchmark systems completed,
- DFT settings table,
- convergence/failure table,
- at least one uncertainty/ranking table,
- one active-learning vs random comparison,
- one newwalk pathway result,
- short technical report,
- limitations clearly written.

## 2-Week Execution Plan

### Days 1–2

Prepare benchmark folder structure and select starting geometries for:

- ferrocene,
- $\text{Ni}(\text{CO})_4$ ,
- $\text{Cr}(\text{CO})_6$ ,
- $\text{Fe}(\text{CO})_5$ .

Create metadata table with charge, spin, geometry source, and DFT settings.

### Days 3–5

Run initial single-point or relaxation calculations for each optimized/reference geometry. Parse outputs with ActiStruct. Confirm that convergence, energy, force, and metadata are correctly recorded.

### Days 6–8

Generate local perturbation candidates around each complex. Start with 10–20 candidates per complex, not hundreds.

### Days 9–10

Train a small baseline model per complex family or a grouped model if enough data exists. Add uncertainty estimate and active-learning ranking.

### Days 11–12

Run the top ActiStruct-selected candidates and compare against random or grid-selected candidates.

### Days 13–14

Run one newwalk pathway only: ferrocene Cp rotation or  $\text{Fe}(\text{CO})_5$  Berry pseudorotation. Prepare a concise technical report.

## Final Mentor Rule

A small, honest, reproducible benchmark is stronger than a large, noisy, overclaimed project.

The goal is not to show that the tools are perfect. The goal is to show that you understand how to build trustworthy DFT + ML workflows for difficult transition-metal chemistry.